

p-values

1. The intensity of light reflected by an object is measured. Suppose there are two types of possible objects, A and B. If the object is of type A, the measurement is normally distributed with mean 100 and standard deviation 25. If the object is of type B, the measurement is normally distributed with mean 125 and standard deviation 25. Researchers plan to take a single measurement X .

Suppose the testing procedure δ has been formulated that declares the object to be of type B if $X > 125$.

- (a) Write out the hypotheses of this test. What is the implied parameter space Ω ? Are the hypotheses simple or composite?
 - (b) What is the size of this test δ ?
 - (c) What is the power (not power function) of this test δ ?
 - (d) Suppose we observe $X = 120$. What is the p -value?
2. Suppose $X \sim \text{Unif}[0, \theta]$. We'd like to test the following hypotheses:

$$H_0 : \theta \leq 1 \quad \text{vs.} \quad H_1 : \theta > 1$$

Let δ_c be the procedure that rejects H_0 if $X \geq c$ for some $c > 0$. It might useful to

recall that if $X \sim \text{Unif}[a, b]$, then $F_X(x) = \begin{cases} 0 & x < a \\ \frac{x-a}{b-a} & a \leq x \leq b \\ 1 & x > b \end{cases}$

- (a) For each possible value of $X = x$, find the form of the p -value if x is observed.
- (b) Briefly interpret your p -value from (a), and discuss why the p -value “makes sense”.